

$$f(z,t) = f_s(z,t) + f_a(z,t) + f_d(z,t)$$

f_s := stem cells per unit length

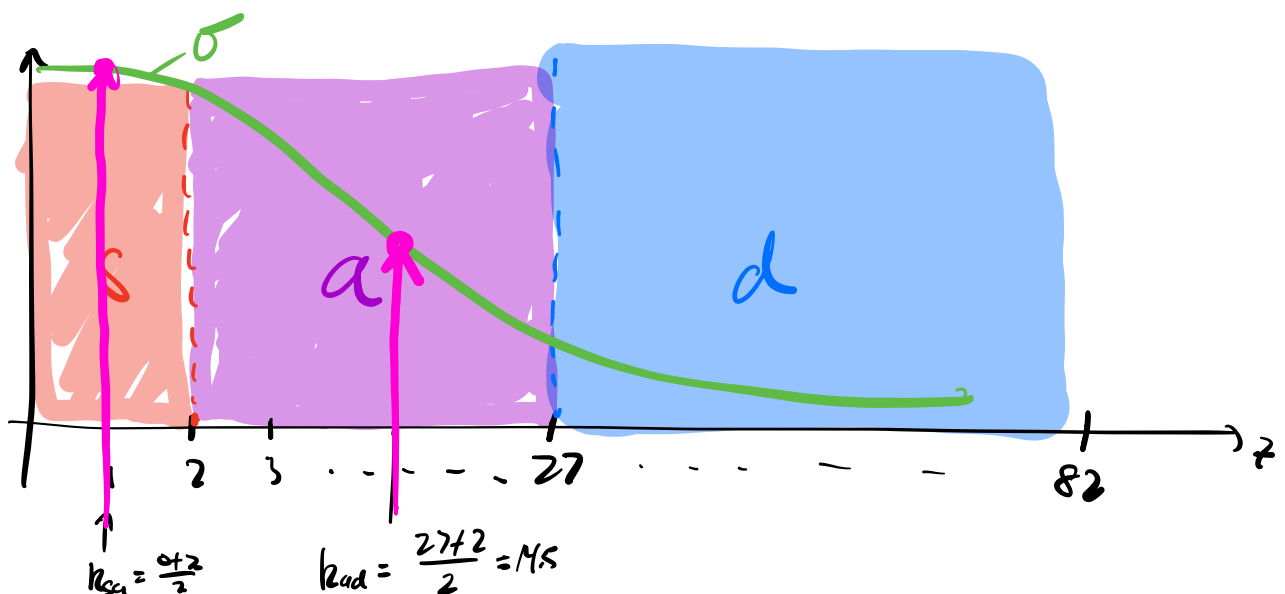
f_a := transient amplifying cells per unit length

f_d := differentiated cells per unit length

$$\frac{df_s}{dt} = -f_{sa}(\sigma) f_s$$

$$\frac{df_a}{dt} = f_{sa}(\sigma) f_s - f_{ad}(\sigma) f_a$$

$$\frac{df_d}{dt} = f_{ad}(\sigma) f_a$$



$$f_{su}(\sigma) = \frac{V_{su}}{1 + \left(\frac{\sigma}{k_{su}}\right)^n}, \quad f_{ad}(\sigma) = \frac{V_{ad}}{1 + \left(\frac{\sigma}{k_{ad}}\right)^n}$$

OR

$g(\sigma, z, t)$ = number of cells per unit length with identical state σ at time t .

$$\sigma \in [0, \infty)$$

$\sigma_s \rightarrow$ stemness threshold for stem cells.

$\sigma_a \rightarrow$ stemness threshold for TA cells

$$n_s(z, t) = \text{density of stem cells} = \int_{\sigma_s}^{\infty} g(\sigma, z, t) d\sigma$$

$$n_a(z, t) = \text{density of TA cells} = \int_{\sigma_a}^{\sigma_s} g(\sigma, z, t) d\sigma$$

$$n_d(z, t) = \text{density of diff cells} = \int_0^{\sigma_a} g(\sigma, z, t) d\sigma$$

$$\frac{\partial g}{\partial t} = \frac{\partial}{\partial t} (g(\sigma(t), z, t)) = \frac{\partial g}{\partial t} + \frac{\partial g}{\partial \sigma} \cdot \frac{d\sigma}{dt}$$